

Estimating Heterogeneous Customer Responses to Pricing Interventions Using Uplift Modelling: A Synthetic Case Study

Independent Research Project

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Abstract

Full-service maintenance contracts are widely used in industrial service operations, yet determining effective pricing strategies remains challenging due to uncertainty in customer behaviour and heterogeneous price sensitivity. While predictive machine learning models can estimate the likelihood of contract acceptance, they do not directly quantify how alternative pricing decisions influence customer responses.

This study investigates the use of uplift modelling to estimate heterogeneous treatment effects of a discount on customer acceptance of service contract offers. A synthetic dataset representing customers with varying operational characteristics was generated to simulate pricing decisions in a controlled environment. A logistic regression model was first developed as a predictive baseline, followed by a T-Learner framework based on Random Forest classifiers to estimate individual treatment effects.

The results indicate that offering a discount increases the average probability of contract acceptance by approximately 10 percentage points while revealing substantial variation across customer segments. Small customers exhibit considerably higher responsiveness to discounts than medium and large customers, suggesting that targeted pricing policies may outperform uniform discounting strategies.

The proposed framework achieved a baseline predictive AUC of **0.666** and estimated an average treatment effect of approximately **10.8 percentage points**, while identifying substantial heterogeneity across customer segments. Although based on synthetic data, the study demonstrates how treatment-effect modelling can support pricing decisions by identifying customers who are most likely to respond to pricing interventions.

Keywords: Uplift Modelling, Causal Machine Learning, Pricing Analytics, Service Contracts, Treatment Effect Estimation, Operations Management.

1. Introduction

Pricing decisions are central to many operational and service-management problems. In industries where equipment maintenance is offered through long-term service contracts, organisations must determine prices that remain attractive to customers while ensuring economic sustainability. Traditional pricing approaches often rely either on estimating expected maintenance costs or on predicting customer willingness to pay. However, these

approaches primarily focus on prediction and provide limited insight into how customers would respond under alternative pricing strategies.

Recent advances in causal machine learning offer a complementary perspective by estimating treatment effects rather than only predicting outcomes. Instead of asking whether a customer is likely to accept a contract, treatment-effect models aim to estimate how the probability of acceptance changes when a pricing intervention—such as offering a discount—is applied. This distinction enables more informed decision-making by identifying customers who are most responsive to specific pricing actions.

The objective of this study is to investigate whether uplift modelling can identify heterogeneous customer responses to pricing interventions in a simplified service-contract setting. Because suitable real-world data were unavailable, a synthetic dataset was constructed to represent customers with different operational characteristics, including machine age, equipment utilisation, previous failures, contract duration, and company size. Although synthetic data cannot reproduce all aspects of industrial decision-making, they provide a controlled environment for demonstrating methodological concepts and analysing model behaviour.

The study follows a two-stage modelling approach. First, a conventional predictive model estimates the probability of contract acceptance. Second, a T-Learner framework estimates individual treatment effects by comparing predicted acceptance probabilities under alternative pricing scenarios. The analysis evaluates both the average impact of discounting and the heterogeneity of customer responses across different customer segments.

The contribution of this work is methodological rather than empirical. Rather than proposing a production-ready pricing model, the study demonstrates an end-to-end workflow for treatment-effect estimation, including synthetic data generation, predictive modelling, uplift estimation, and customer segmentation. The proposed framework illustrates how causal machine learning techniques can support pricing decisions and provides a foundation for future research using real-world operational data.

2. Methodology

2.1 Problem Formulation

The objective of this study is to investigate how customer acceptance of full-service maintenance contracts changes under alternative pricing strategies. Rather than predicting whether a customer will accept a contract, the study focuses on estimating how the probability of acceptance changes when a pricing intervention is applied.

Formally, each customer is characterised by a set of observable attributes

$$X = (x_1, x_2, \dots, x_p),$$

including machine characteristics, historical maintenance information, contract duration, and customer segment.

The treatment variable

$$T \in \{0,1\}$$

indicates whether a customer receives a 10% discount on the offered contract price.

The outcome variable

$$Y \in \{0,1\}$$

represents whether the customer accepts the maintenance contract.

The primary objective is to estimate the individual treatment effect

$$ITE(x) = P(Y = 1 | T = 1, X = x) - P(Y = 1 | T = 0, X = x),$$

which measures the expected increase in acceptance probability resulting from the discount.

2.2 Synthetic Data Generation

Since no suitable real-world pricing dataset was available, a synthetic dataset was generated to simulate customer responses under controlled conditions.

The dataset contains **5,000** simulated customers described by five explanatory variables:

Table 1. Variables included in the synthetic service contract dataset

Variable	Description
Machine age	Age of the serviced equipment (years)
Annual usage	Annual operating hours
Customer size	Small, Medium or Large
Contract duration	Length of the maintenance contract (1–5 years)
Previous failures	Number of historical equipment failures

Each customer was randomly assigned either to a control group or to a treatment group receiving a **10% price discount**.

Customer acceptance probabilities were generated using a logistic model in which acceptance depends on both operational characteristics and customer segment. The treatment effect was intentionally designed to be heterogeneous, allowing different customer groups to exhibit different levels of price sensitivity. Additional random variation was introduced to simulate unobserved behavioural factors and produce realistic variability in customer responses.

Although synthetic, the generated data capture several characteristics commonly observed in pricing problems, including heterogeneous treatment effects and uncertainty in customer decision-making.

2.3 Predictive Modelling

As a predictive baseline, a Logistic Regression classifier was trained to estimate the probability of contract acceptance using all available customer characteristics.

Model performance was evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC), providing a measure of the model's ability to discriminate between accepted and rejected contracts.

The predictive model serves as a benchmark and illustrates the difference between conventional prediction and treatment-effect estimation.

2.4 Treatment Effect Estimation

To estimate heterogeneous treatment effects, a **T-Learner** framework was implemented.

Two independent Random Forest classifiers were trained:

- one using observations from customers who received a discount;
- one using observations from customers who did not receive a discount.

For every customer, the probability of contract acceptance was estimated under both scenarios.

The estimated treatment effect was computed as the difference between these two predicted probabilities:

$$\widehat{ITE}(x) = \hat{P}(Y = 1 \mid T = 1, X = x) - \hat{P}(Y = 1 \mid T = 0, X = x).$$

This approach enables the identification of customers who are expected to benefit most from a pricing intervention, supporting targeted pricing strategies instead of uniform discount policies.

Finally, the estimated individual treatment effects were analysed at both the customer and segment levels. Summary statistics, customer ranking based on estimated uplift, and graphical visualisations were used to interpret the distribution of treatment effects and identify groups expected to benefit most from pricing interventions.

3. Results

3.1 Predictive Performance

The baseline Logistic Regression model achieved an Area Under the Receiver Operating Characteristic Curve (AUC) of **0.666**, indicating a moderate ability to distinguish between customers who accept and reject maintenance contract offers. Although the predictive model captures general behavioural patterns, it does not estimate how customer decisions change under alternative pricing strategies.

To address this limitation, treatment-effect modelling was applied to estimate the causal impact of offering a discount on contract acceptance.

Table 2 summarises the principal quantitative results.

Table 2. Summary of modelling results

Metric	Value
Number of customers	5,000
Baseline model AUC	0.666
Average Estimated Treatment Effect	0.108
Maximum Estimated Treatment Effect	0.258

The predictive performance indicates that the baseline model captures meaningful behavioural patterns, while the treatment-effect model provides additional information by estimating counterfactual customer responses under alternative pricing scenarios.

The estimated Average Treatment Effect (ATE) suggests that offering a **10% discount** increases the probability of contract acceptance by approximately **10.8 percentage points** on average across the simulated customer population.

3.2 Distribution of Treatment Effects

Figure 1 illustrates the distribution of estimated individual treatment effects.

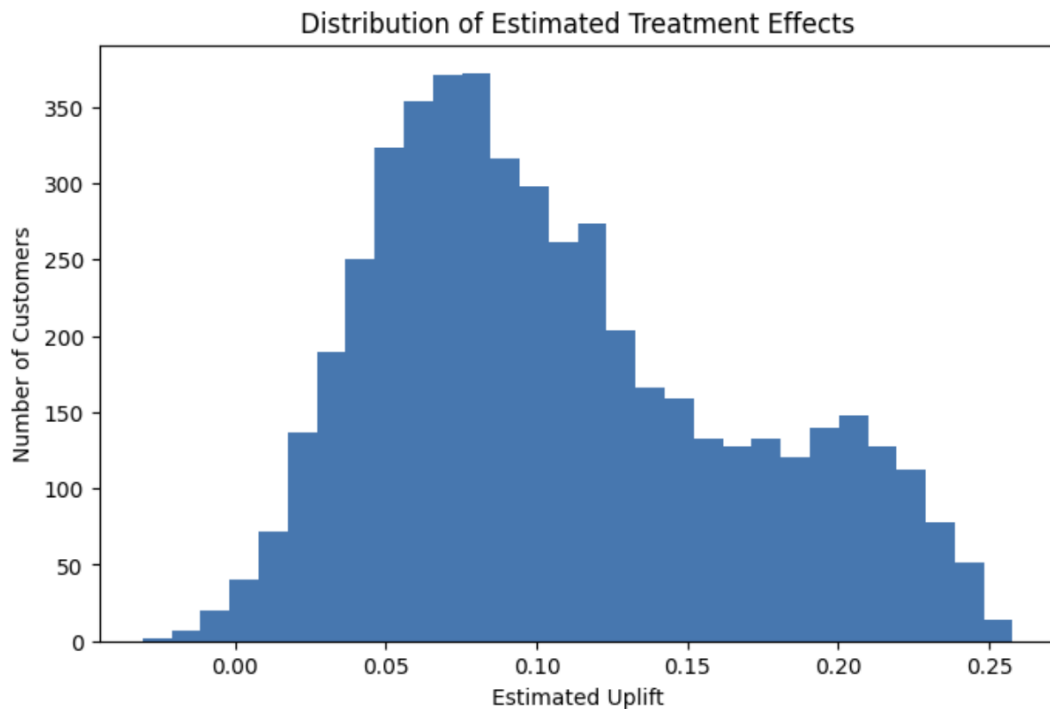


Figure 1. Distribution of estimated treatment effects

The estimated treatment effects are distributed over a relatively wide range, indicating considerable heterogeneity in customer responses. While many customers exhibit modest improvements in acceptance probability, a smaller group is predicted to respond substantially more strongly to the pricing intervention.

This variation demonstrates that customers should not be treated as equally price-sensitive. Instead, estimating individual treatment effects enables organisations to identify customers who are expected to benefit most from targeted discounts.

The positively skewed distribution further suggests that a relatively small proportion of customers accounts for the largest expected gains from discounting, reinforcing the value of targeted rather than uniform pricing strategies.

3.3 Customer Segmentation

Average treatment effects were further analysed across customer segments.

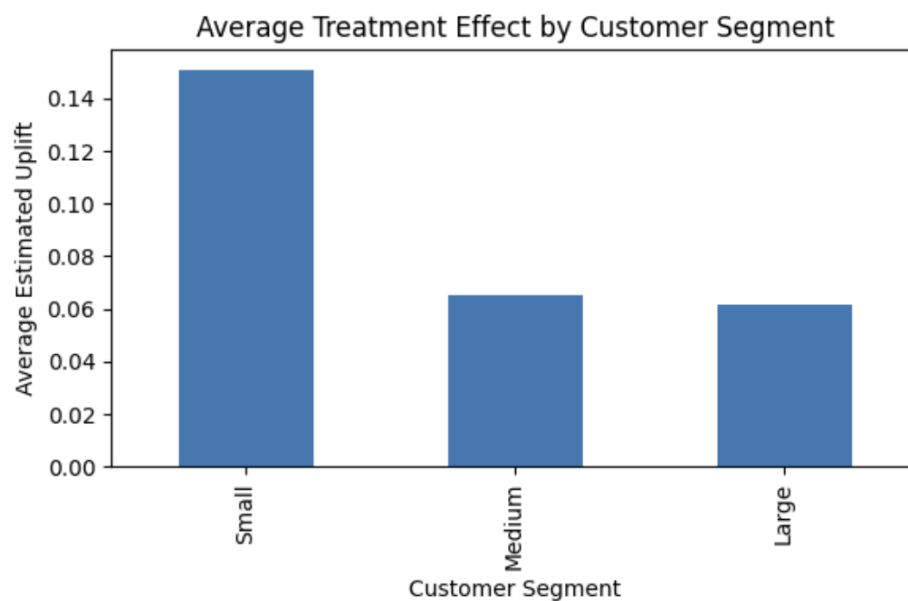


Figure 2. Average treatment effect by customer segment

The results reveal clear differences in price responsiveness between customer groups. Small customers exhibit the highest estimated treatment effect, with an average increase in acceptance probability of approximately **15 percentage points**. Medium and large customers demonstrate considerably smaller responses, with average treatment effects of approximately **6–7 percentage points**.

These findings indicate that customer size is associated with substantial differences in price sensitivity within the simulated environment.

The results suggest that customer segment alone explains a substantial share of treatment-effect heterogeneity in the simulated environment, although additional operational characteristics also contribute to individual differences in predicted uplift.

Although the synthetic data generation process intentionally introduced heterogeneous treatment effects, the modelling framework successfully recovered these differences, demonstrating its ability to identify customer groups that are most responsive to pricing interventions.

4. Discussion

The results illustrate the distinction between predictive modelling and treatment-effect estimation. A conventional predictive model estimates the likelihood that a customer will accept a contract, whereas uplift modelling estimates how that likelihood changes under an alternative pricing decision.

From a methodological perspective, the study illustrates the distinction between predictive and causal machine learning. While predictive models estimate expected outcomes under observed conditions, treatment-effect models explicitly evaluate counterfactual scenarios by comparing predicted outcomes under different interventions. This capability is particularly valuable in pricing analytics, where decision-makers are interested not only in predicting customer behaviour but also in estimating how that behaviour may change under alternative pricing policies.

From a managerial perspective, this distinction is important. Applying uniform discounts to all customers may increase acceptance rates but can also reduce profitability by offering unnecessary price reductions to customers who would have accepted the original offer. Treatment-effect models instead support more targeted pricing strategies by identifying customers who are expected to respond most strongly to discounts.

In this study, the strongest treatment effects were observed among small customers, suggesting that pricing interventions may generate greater value when directed toward customer groups with higher price sensitivity. Although this pattern reflects the assumptions used during synthetic data generation, it demonstrates how heterogeneous treatment-effect estimation can support operational decision-making.

The study also illustrates the practical applicability of uplift modelling in pricing analytics. Even in a simplified setting, treatment-effect estimation provides richer information than conventional prediction by enabling the comparison of alternative pricing scenarios for individual customers.

5. Limitations

Several limitations should be acknowledged.

First, the study relies entirely on synthetic data rather than observations collected from industrial service operations. Consequently, the estimated treatment effects should be interpreted as a demonstration of methodology rather than empirical evidence.

Second, customer behaviour was generated using predefined probabilistic relationships. Real-world purchasing decisions are influenced by many additional factors, including market conditions, competitive pricing, customer relationships, and contractual negotiations, which were intentionally omitted to maintain a tractable experimental setting.

Finally, the study employs a T-Learner framework using Random Forest models. Although this approach provides an intuitive introduction to heterogeneous treatment-effect estimation, more advanced causal machine learning methods, such as X-Learners, Doubly Robust Learners, or Causal Forests, could potentially improve estimation accuracy in practical applications.

Despite these limitations, synthetic experiments remain valuable for evaluating methodological frameworks under controlled conditions before applying them to observational industrial datasets.

6. Conclusion

This study investigated the application of uplift modelling to estimate heterogeneous customer responses to pricing interventions in maintenance service contracts.

Using a synthetic dataset representing operational customer characteristics, a predictive baseline model and a T-Learner treatment-effect model were developed to evaluate the impact of a 10% discount on contract acceptance. The results demonstrated a positive average treatment effect while revealing substantial heterogeneity across customer segments.

Although the study does not provide empirical evidence for industrial pricing decisions, it demonstrates a complete analytical workflow for heterogeneous treatment-effect estimation in service-contract settings. The proposed framework can be readily extended to real-world maintenance data and more advanced causal machine learning algorithms, providing a basis for future research in data-driven pricing and operations management.

7. References

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